

OFFICIAL ARCHITECTURE & OPERATIONAL MANUAL

GETRA Backend

Complete Technical Documentation, API Catalog, Database & PostGIS Foundation, Supabase Migration Runbook, Docker Operations & Security Handover

Project	GETRA (Gerakan Transit Ramah Anak)
Document Version	1.0.0 (Final Backend Baseline)
Release Phase	Extra Phase Final – Documentation & Handover
Runtime Stack	Next.js 16.3.1 Standalone / Node 22 LTS
Database Platform	Supabase PostgreSQL 15+ with PostGIS
Supabase Project Ref	sesakxnjaphrxqxllqjm
Audit Status	VERIFIED & 100% AUDITED AGAINST REPOSITORY
Date	2026-08-17 (Asia/Jakarta)

GETRA Backend Complete Technical Documentation

Architecture, API Catalog, Database & PostGIS Foundation, Docker Operations, Security & Operational Handover

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Document Version: 1.0.0 (Final Backend Baseline)

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Target Environment: Node.js 22 LTS / Next.js 16.3.1 Standalone / Supabase PostgreSQL & PostGIS Remote

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1. Executive Summary

GETRA (Gerakan Transit Ramah Anak) Backend is a high-performance, modular geospatial and transit data platform built on **Next.js 16 (App Router)** and **Supabase (PostgreSQL with PostGIS)**. The backend acts as the core transactional, spatial, and integration engine supporting transit node mapping, study area demarcations, corridor routing foundations, UMKM business directory integration, and role-based commuter services.

Core Capabilities Summary:

- **Architecture:** Layered Clean Architecture (HTTP Handler $\rightarrow$ Security Middleware $\rightarrow$ Zod Validation $\rightarrow$ Service $\rightarrow$ Repository $\rightarrow$ PostGIS).
 - **Database & PostGIS:** 6 normalized PostgreSQL tables with strict constraints, EPSG:4326 spatial geometry types (Point, MultiLineString, MultiPolygon), GiST indexing, and server-side RPC functions for bounded bounding-box and proximity queries.
 - **Authentication & Authorization:** Stateless Supabase JWT Bearer token authentication with 4 roles (**COMMUTER** , **UMKM** , **COMMUNITY** , **ADMIN**), Row Level Security (RLS) enforcement, and column-level permission boundaries.
 - **Security Hardening:** Strict origin CORS allowlist, granular rate limiting, bounded streaming JSON parsing, secure HTTP headers (CSP, HSTS, X-Frame-Options, etc.), request IDs, and non-root Docker execution.
 - **Integrations:** Extensible MAPID / External Data Adapter foundation featuring fixture-based testing, dry-run simulation, and data provenance tracking.
 - **Operational Readiness:** Fully dockerized multi-stage standalone runtime with automated health checking and 24/7 PM2 process management support.
-

2. GETRA Backend Overview

The GETRA backend provides secure, bounded RESTful endpoints for web and mobile frontends. It is designed around the principle of **strict contract safety**: no unverified external dependencies are invoked in production, all queries are bounded to prevent denial-of-service, and sensitive credentials (such as service-role keys or database passwords) are strictly quarantined from public runtime bundles.

System Verification & Implementation Status:

- **Overall Status:** FOUNDATION IMPLEMENTED – ACCURACY AUDITED
 - **Real Production Data:** PENDING PRODUCTION INGESTION
 - **Real MAPID Connection:** FOUNDATION ONLY – OFFICIAL CONTRACT REQUIRED
 - **Pedestrian Graph Routing:** FOUNDATION ONLY – PRODUCTION GRAPH REQUIRED
-

3. Technology Stack

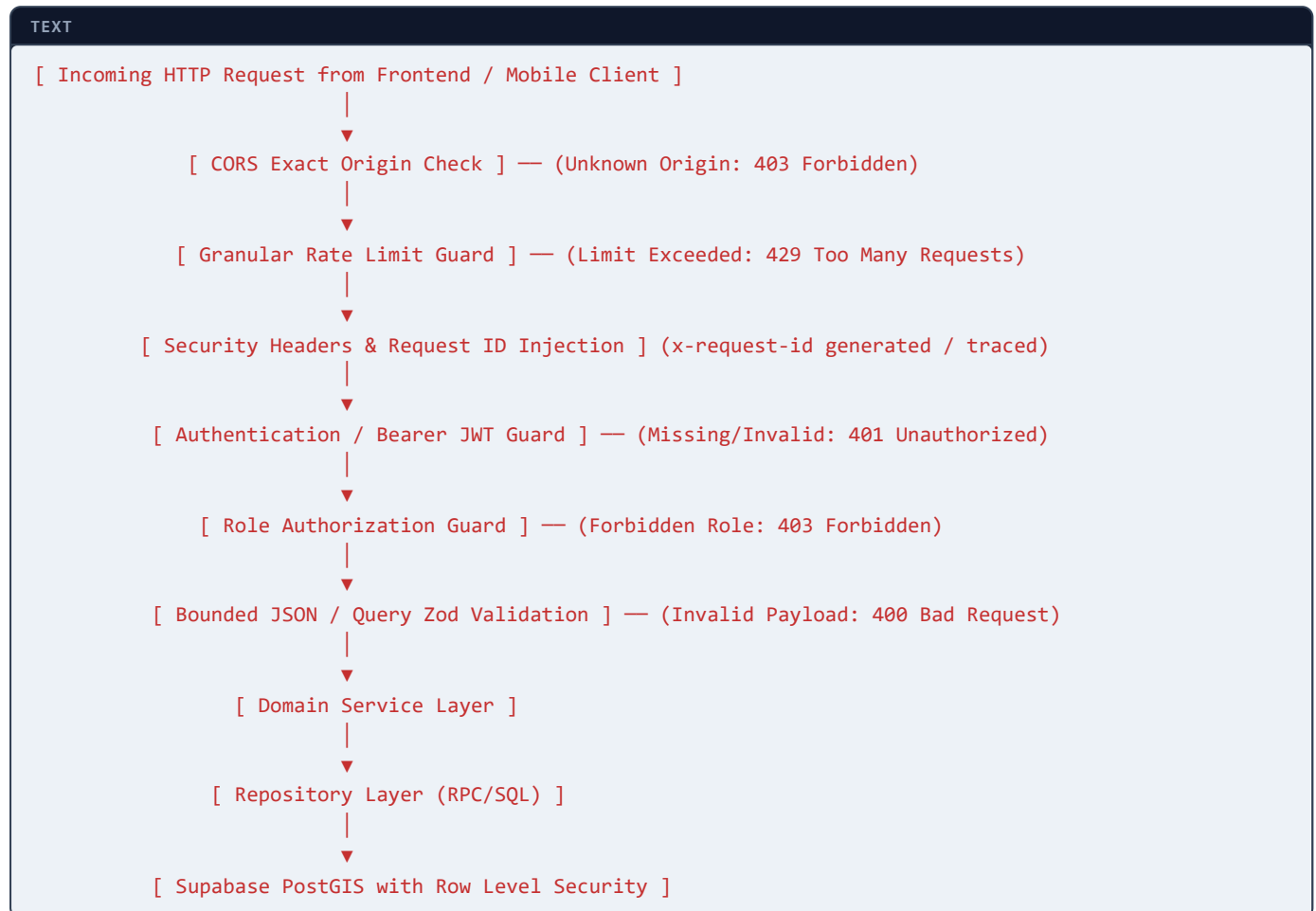
The GETRA backend stack is strictly documented from `package.json`, source code, and configuration files:

Layer / Category	Technology / Library	Version	Purpose in GETRA
Language	TypeScript	<code>^6.0.3</code>	Type-safe backend domain models, DTOs, and schemas
Framework	Next.js	<code>^16.3.1</code>	App Router API Route Handlers and Standalone Server
Runtime Core	React / React DOM	<code>^19.2.8</code>	Core Next.js 16 engine peer dependencies
Server Boundary	<code>server-only</code>	<code>^0.0.1</code>	Prevents server modules from leaking into client bundles
Validation	Zod	<code>^4.4.3</code>	Runtime schema validation for requests, configs, and payloads
Database SDK	<code>@supabase/supabase-js</code>	<code>^2.112.3</code>	Supabase PostgreSQL client and authentication provider
Database Engine	PostgreSQL + PostGIS	Remote (Supabase)	Spatial data storage, GiST indexing, and spatial calculations
Container Engine	Docker & Docker Compose	Node 22-slim	Standalone containerization and reproducible deployment
Process Manager	PM2	<code>^7.0.3</code>	24/7 host process monitoring and automatic crash restarts
Testing Engine	Vitest	<code>^4.1.10</code>	Unit, integration, and security test runner
Linter / Quality	ESLint / eslint-config-next	<code>^9.39.5</code> / <code>^16.3.1</code>	Static code analysis and style verification
CLI Tooling	Supabase CLI	<code>^2.114.0</code>	Local database management and migration tooling

4. System Architecture

GETRA implements a strict, multi-tiered architecture that separates concerns between transport protocol, security guards, input validation, domain business logic, data persistence, and database-level RLS.

4.1 Standard Request Flow



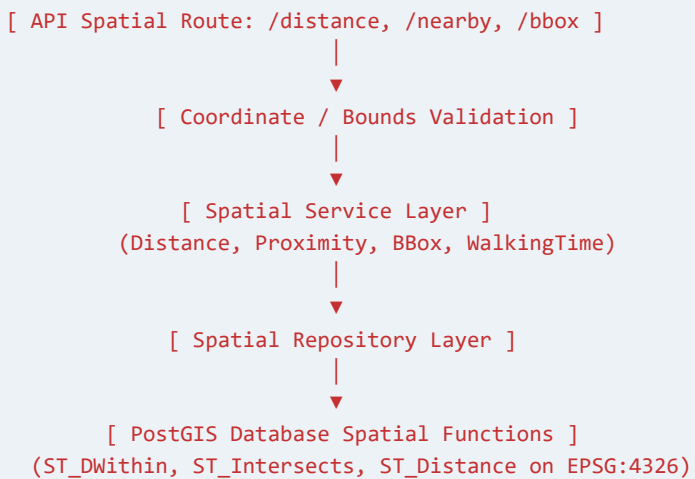
4.2 External Provider / MAPID Adapter Pipeline

TEXT



4.3 Spatial Engine Flow

TEXT



5. Project Folder Structure

The repository structure reflects Clean Architecture with strict separation of routes, schemas, domain modules, repositories, and migration files:

TEXT

```

getra-backend/
├── app/
│   ├── api/
│   │   ├── auth/
│   │   │   ├── login/route.ts
│   │   │   ├── logout/route.ts
│   │   │   ├── me/route.ts
│   │   │   └── register/route.ts
│   │   ├── health/
│   │   │   └── route.ts
│   │   ├── profile/
│   │   │   └── route.ts
│   │   ├── spatial/
│   │   │   ├── bbox/route.ts
│   │   │   ├── distance/route.ts
│   │   │   └── nearby/route.ts
│   │   └── v1/
│   └── src/
│       ├── config/
│       ├── integrations/
│       │   ├── core/
│       │   └── mapid/
│       ├── lib/
│       │   ├── api-security/
│       │   ├── env/
│       │   ├── errors/
│       │   ├── http/
│       │   ├── logger/
│       │   ├── spatial/
│       │   ├── supabase/
│       │   ├── api-logger.ts
│       │   ├── api-response.ts
│       │   ├── auth.ts
│       │   ├── pagination.ts
│       │   ├── query-parser.ts
│       │   ├── rate-limit.ts
│       │   ├── request-body.ts
│       │   ├── request-id.ts
│       │   └── validation.ts
│       ├── mappers/
│       ├── modules/
│       │   ├── spatial/
│       │   └── spatial-import/
│       ├── repositories/
│       ├── schemas/
│       ├── services/
│       └── types/
├── supabase/
│   ├── migrations/
│   ├── tests/database/
│   ├── config.toml
│   └── seed.sql
├── tests/
│   ├── fixtures/
│   ├── integration/
│   └── unit/
├── deployment/
├── docs/
│   ├── changes/
│   └── database/

```

Next.js App Router root
 # API route handlers
 # Authentication endpoints
 # POST /api/auth/login, OPTIONS
 # POST /api/auth/logout, OPTIONS
 # GET /api/auth/me, OPTIONS
 # POST /api/auth/register, OPTIONS
 # Health & readiness probe
 # GET /api/health, OPTIONS
 # Profile management
 # GET, PATCH /api/profile, OPTIONS
 # Geospatial endpoints
 # GET /api/spatial/bbox, OPTIONS
 # POST /api/spatial/distance, OPTIONS
 # GET /api/spatial/nearby, OPTIONS
 # Versioned endpoint placeholder (.gitkeep)
 # Core backend source code
 # Application bootstrap & runtime config
 # External data integrations & adapters
 # Common external provider contracts & errors
 # MAPID adapter, client, mapper, normalizer
 # Shared utility libraries & security helpers
 # CORS, security headers, endpoint policies
 # Type-safe environment validation
 # ApplicationError hierarchy & error codes
 # Timeout fetch & abort controller helpers
 # Sanitized structured logging helpers
 # Spatial parsing, GeoJSON mappers & SRID helpers
 # Server & request-scoped Supabase client factories
 # withApiLogger request wrapper
 # Standard JSON envelope creators
 # Authentication & role guard utilities
 # List pagination helpers
 # Query parameter parsers
 # In-memory fixed-window rate limiter
 # Bounded stream JSON body reader
 # UUID request tracer
 # Zod request body & query validators
 # Domain, geometry, and profile data mappers
 # Specialized business logic domain modules
 # PostGIS spatial calculation & query services
 # Spatial import contracts & normalization
 # Data access layer interfacing with Supabase
 # Central Zod validation schemas
 # Domain service layer
 # TypeScript interfaces and entity types
 # Supabase database configuration
 # PostgreSQL migrations (source of truth)
 # pgTAP database tests
 # Local Supabase CLI configuration
 # Synthetic DEV/TEST seed fixture data
 # Comprehensive Vitest test suite
 # Synthetic GeoJSON and MAPID test fixtures
 # API route & security integration tests
 # Unit tests for models, services & repositories
 # Deployment environment guidelines
 # Technical documentation & change logs
 # Phase change logs (Phase 4, 6, 7, Extra, Final)
 # Database architecture documentation

```
| └─ final/                # Complete master technical documentation & PDF
| Dockerfile              # 3-stage standalone Node 22 production Dockerfile
| docker-compose.yml      # Base Docker Compose specification
| docker-compose.prod.yml # Production Docker Compose security overrides
| ecosystem.config.cjs    # PM2 process manager configuration
| next.config.ts          # Next.js configuration (standalone output)
| tsconfig.json           # TypeScript compiler configuration
| package.json            # Project dependencies and script definitions
```

6. Authentication System

GETRA utilizes **Supabase Auth** as the underlying authentication provider, utilizing JSON Web Tokens (JWT) transmitted via HTTP `Authorization: Bearer <token>` headers.

6.1 Authentication Lifecycle

1. Registration (`POST /api/auth/register`):

- Creates a new user in `auth.users`.
- Trigger `public.handle_new_user()` automatically creates a corresponding record in `public.profiles`.
- **Security Rule:** Public registration strictly permits `COMMUTER`, `UMKM`, or `COMMUNITY` roles. Requesting `ADMIN` triggers an immediate `403 Forbidden` error.

2. Login (`POST /api/auth/login`):

- Authenticates user credentials via `supabase.auth.signInWithPassword()`.
- Returns `access_token`, user identifier, email, and associated profile summary (`display_name`, `role`).

3. Session Verification (`GET /api/auth/me`):

- Validates bearer token against Supabase Auth.
- Fetches user profile using a request-scoped Supabase client subject to RLS.

4. Logout (`POST /api/auth/logout`):

- Acknowledges stateless logout (`token_disposition: "client_discard_required"`).
- Frontend client must immediately purge the token from local storage / memory.

6.2 Bearer Token Extraction & Validation

Bearer tokens are bounded to a maximum of 8,192 characters and validated against the format `^Bearer ([^\s]+)$`. Unauthenticated or malformed requests are rejected with a standardized `UNAUTHORIZED` error (HTTP 401).

7. User Profiles & Roles

GETRA defines four explicit user roles via the PostgreSQL enum type `public.user_role` :

Role Enum	Description	Intended Platform Capabilities
<code>COMMUTER</code>	General public transit user	Public transit search, nearby stops, bookmarks, walking calculations
<code>UMKM</code>	Verified merchant / business owner	Manage owned UMKM profile, location, business information
<code>COMMUNITY</code>	Community leader / local group	View community transit initiatives and report transit observations
<code>ADMIN</code>	System administrator	Manage all spatial sources, reference nodes, corridors, and all profiles

7.1 Role Authorization Matrix

Endpoint	Public	COMMUTER	UMKM	COMMUNITY	ADMIN
<code>GET /api/health</code>	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$
<code>POST /api/auth/login</code>	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$
<code>POST /api/auth/register</code>	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\times\$ (Blocked)
<code>POST /api/auth/logout</code>	\$\times\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$
<code>GET /api/auth/me</code>	\$\times\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$
<code>GET /api/profile</code>	\$\times\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$
<code>PATCH /api/profile</code>	\$\times\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$
<code>POST /api/spatial/distance</code>	\$\times\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$
<code>GET /api/spatial/nearby</code>	\$\times\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$
<code>GET /api/spatial/bbox</code>	\$\times\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$

7.2 Administrator Provisioning Security Policy

CAUTION

Admin Role Protection: The `ADMIN` role cannot be provisioned through public registration. The trigger `public.handle_new_user()` will raise exception `ROLE_NOT_ALLOWED` if `role = 'ADMIN'` is provided in metadata. Admin privileges must be assigned manually by a database administrator via direct SQL or a secure administrative migration script.

8. Row Level Security (RLS) & Authorization

Row Level Security is enabled on every table in the database. Client queries are executed with the user's JWT, ensuring that database-level authorization is enforced by PostgreSQL even if application-level checks fail.

8.1 Complete RLS Policy Table

Table Name	RLS Enabled	Allowed Roles	Operation	Policy Definition / Owner Rule
public.profiles	YES	authenticated	SELECT	auth.uid() = id (Read own profile)
public.profiles	YES	authenticated	UPDATE	auth.uid() = id (Update own profile)
public.profiles	YES	authenticated	SELECT	private.is_admin() (Admins read all)
public.profiles	YES	authenticated	UPDATE	private.is_admin() (Admins update all)
public.spatial_sources	YES	authenticated	SELECT	true (All authenticated can read)
public.spatial_sources	YES	authenticated	ALL	private.is_admin() (Admins manage)
public.study_areas	YES	authenticated	SELECT	true (All authenticated can read)
public.study_areas	YES	authenticated	ALL	private.is_admin() (Admins manage)
public.transport_corridors	YES	authenticated	SELECT	true (All authenticated can read)
public.transport_corridors	YES	authenticated	ALL	private.is_admin() (Admins manage)
public.transport_nodes	YES	authenticated	SELECT	true (All authenticated can read)
public.transport_nodes	YES	authenticated	ALL	private.is_admin() (Admins manage)
public.umkm_profiles	YES	authenticated	SELECT	owner_id = auth.uid() (Owner read)
public.umkm_profiles	YES	authenticated	INSERT	owner_id = auth.uid() (Owner create)
public.umkm_profiles	YES	authenticated	UPDATE	owner_id = auth.uid() (Owner update)
public.umkm_profiles	YES	authenticated	ALL	private.is_admin() (Admins manage all)

8.2 Column-Level Grants & Privilege Hardening

To prevent privilege escalation, column-level permission restrictions are enforced in PostgreSQL:

- **profiles Table:** Authenticated users only have `UPDATE` privileges on `(display_name, avatar_url)`. Role assignment cannot be modified by user updates.
- **umkm_profiles Table:** Authenticated users only have `INSERT` privileges on `(owner_id, business_name, category, description, address, geometry)`. System provenance columns (`source_id`, `source_record_id`, `data_version`, `validation_status`, `retrieved_at`, `validated_at`, `metadata`) cannot be forged on insert.
- **Trigger `private.enforce_umkm_trusted_fields_update()`**: Blocks non-admin users from altering provenance and ownership metadata during updates.

9. API Architecture & Response Contracts

Every API response from GETRA follows a strict, predictable JSON envelope format.

9.1 Standard Success Envelope (HTTP 200 / 201)

JSON

```
{
  "success": true,
  "data": { ... },
  "request_id": "4d983e20-7ad1-49b8-bc88-34827fb70588"
}
```

9.2 Standard Paginated List Envelope (HTTP 200)

JSON

```
{
  "success": true,
  "data": [ ... ],
  "meta": {
    "page": 1,
    "limit": 20,
    "total": 100,
    "total_pages": 5
  },
  "request_id": "4d983e20-7ad1-49b8-bc88-34827fb70588"
}
```

9.3 Standard Error Envelope (HTTP 4xx / 5xx)

JSON

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Request validation failed",
    "retryable": false
  },
  "request_id": "4d983e20-7ad1-49b8-bc88-34827fb70588"
}
```

9.4 Global Response Headers

All API responses automatically include:

- **x-request-id** : Tracing identifier for telemetry and error tracking.
- **Cache-Control: no-store** : Prevents caching of authenticated and dynamic spatial responses.
- **Retry-After** : Included on HTTP 429 (**RATE_LIMIT_EXCEEDED**) with wait duration in seconds.

10. Database Architecture & Schema

The GETRA database schema consists of 6 tables created and managed via versioned Supabase migrations.

MERMAID

```

erDiagram
    auth_users ||--|| profiles : "references (1:1)"
    profiles ||--o{ umkm_profiles : "owns (1:N)"
    spatial_sources ||--o{ study_areas : "lineage (1:N)"
    spatial_sources ||--o{ transport_corridors : "lineage (1:N)"
    spatial_sources ||--o{ transport_nodes : "lineage (1:N)"
    spatial_sources ||--o{ umkm_profiles : "lineage (1:N)"
    transport_corridors ||--o{ transport_nodes : "connects (1:N)"

    profiles {
        uuid id PK
        text display_name
        text avatar_url
        user_role role
        timestamptz created_at
        timestamptz updated_at
    }

    spatial_sources {
        uuid id PK
        text source_name
        text source_type
        text description
        jsonb metadata
        timestamptz created_at
        timestamptz updated_at
    }

    study_areas {
        uuid id PK
        uuid source_id FK
        text name
        text description
        geometry geometry
        text source_record_id
        text data_version
        validation_status validation_status
        timestamptz retrieved_at
        timestamptz validated_at
        jsonb metadata
        timestamptz created_at
        timestamptz updated_at
    }

    transport_corridors {
        uuid id PK
        uuid source_id FK
        text name
        text transport_mode
        text description
        geometry geometry
        text source_record_id
        text data_version
        validation_status validation_status
        timestamptz retrieved_at
        timestamptz validated_at
        jsonb metadata
        timestamptz created_at
        timestamptz updated_at
    }

```

```

transport_nodes {
  uuid id PK
  uuid source_id FK
  uuid corridor_id FK
  text name
  text node_type
  text transport_mode
  geometry geometry
  text source_record_id
  text data_version
  validation_status validation_status
  timestampz retrieved_at
  timestampz validated_at
  jsonb metadata
  timestampz created_at
  timestampz updated_at
}

umkm_profiles {
  uuid id PK
  uuid owner_id FK
  uuid source_id FK
  text business_name
  text category
  text description
  text address
  geometry geometry
  text source_record_id
  text data_version
  validation_status validation_status
  timestampz retrieved_at
  timestampz validated_at
  jsonb metadata
  timestampz created_at
  timestampz updated_at
}

```

10.1 Table Specifications

1. `public.profiles`

- **Purpose:** Stores user profile information and role assignments linked directly to Supabase Auth.
- **Primary Key:** `id` (`uuid` , references `auth.users(id)` ON DELETE CASCADE).
- **Columns:** `display_name` (text), `avatar_url` (text nullable), `role` (`user_role` NOT NULL DEFAULT `'COMMUTER'`), `created_at` (timestampz), `updated_at` (timestampz).
- **Indexes:** PK index on `id` .

2. `public.spatial_sources`

- **Purpose:** Lineage catalog for all spatial layers (e.g. MAPID, field surveys, manual entries).
- **Primary Key:** `id` (`uuid` DEFAULT `gen_random_uuid()`).
- **Columns:** `source_name` (text NOT NULL), `source_type` (text NOT NULL: `'external'` , `'survey'` , `'manual'` , `'imported'` , `'system'`), `description` (text nullable), `metadata` (jsonb NOT NULL DEFAULT `'{}'`), `created_at` (timestampz), `updated_at` (timestampz).
- **Indexes:** `idx_spatial_sources_type` (B-tree on `source_type`).

3. `public.study_areas`

- **Purpose:** Boundary polygons representing regional study and transit planning zones.
- **Primary Key:** `id` (`uuid` DEFAULT `gen_random_uuid()`).
- **Geometry:** `geometry(MultiPolygon, 4326)` NOT NULL with `is_valid_wgs84_geometry` check.
- **Lineage & Validation:** `source_id` (FK $\rightarrow$ `spatial_sources.id`), `source_record_id` (text), `data_version` (text), `validation_status` (`validation_status` enum: 'PENDING', 'VALIDATED', 'REJECTED', 'ARCHIVED'), `retrieved_at`, `validated_at`, `metadata` .
- **Indexes:** `idx_study_areas_geometry_gist` (GiST), `idx_study_areas_source_id` (B-tree), `idx_study_areas_source_record_unique` (Unique B-tree on (`source_id`, `source_record_id`)).

4. `public.transport_corridors`

- **Purpose:** Linear route geometries representing public transit lines, corridors, and pathways.
- **Primary Key:** `id` (`uuid` DEFAULT `gen_random_uuid()`).
- **Geometry:** `geometry(MultilineString, 4326)` NOT NULL with `is_valid_wgs84_geometry` check.
- **Columns:** `name` (text NOT NULL), `transport_mode` (text NOT NULL), `description` (text nullable), lineage columns.
- **Indexes:** `idx_transport_corridors_geometry_gist` (GiST), `idx_transport_corridors_source_id` (B-tree), `idx_transport_corridors_mode` (B-tree), `idx_transport_corridors_source_record_unique` (Unique B-tree on (`source_id`, `source_record_id`)).

5. `public.transport_nodes`

- **Purpose:** Point locations for bus stops, train stations, bike shares, and transit access points.
- **Primary Key:** `id` (`uuid` DEFAULT `gen_random_uuid()`).
- **Geometry:** `geometry(Point, 4326)` NOT NULL with `is_valid_wgs84_geometry` check.
- **Columns:** `name` (text NOT NULL), `node_type` (text NOT NULL), `transport_mode` (text NOT NULL), `corridor_id` (FK $\rightarrow$ `transport_corridors.id`), lineage columns.
- **Indexes:** `idx_transport_nodes_geometry_gist` (GiST on geometry), `idx_transport_nodes_geometry_geography_gist` (GiST on (`geometry::geography`)), `idx_transport_nodes_source_id` (B-tree), `idx_transport_nodes_corridor_id` (B-tree), `idx_transport_nodes_mode_type` (B-tree), `idx_transport_nodes_source_record_unique` (Unique B-tree on (`source_id`, `source_record_id`)).

6. `public.umkm_profiles`

- **Purpose:** Registered micro, small, and medium enterprise locations and business directories.
- **Primary Key:** `id` (`uuid` DEFAULT `gen_random_uuid()`).
- **Foreign Keys:** `owner_id` (FK $\rightarrow$ `profiles.id` ON DELETE RESTRICT), `source_id` (FK $\rightarrow$ `spatial_sources.id` ON DELETE SET NULL).
- **Geometry:** `geometry(Point, 4326)` NOT NULL with `is_valid_wgs84_geometry` check.
- **Columns:** `business_name` (text NOT NULL), `category` (text NOT NULL), `description` (text nullable), `address` (text nullable), lineage columns.
- **Indexes:** `idx_umkm_profiles_geometry_gist` (GiST on geometry), `idx_umkm_profiles_geometry_geography_gist` (GiST on (`geometry::geography`)), `idx_umkm_profiles_owner_id` (B-tree), `idx_umkm_profiles_source_id` (B-tree), `idx_umkm_profiles_category` (B-tree), `idx_umkm_profiles_source_record_unique` (Unique B-tree on (`source_id`, `source_record_id`)).

11. PostGIS Foundation

11.1 Spatial Reference & Standards

- **Canonical SRID:** `EPSG:4326` (WGS 84 coordinate system).
- **Coordinates Format:** Standard GeoJSON coordinate order `[longitude, latitude]` for geometry objects, and named `{ longitude, latitude }` objects for API request bodies.
- **Coordinate Bounds Enforced:** Longitude `$[-180.0, 180.0]$, Latitude $[-90.0, 90.0]$.`

11.2 Database Functions & Stored Procedures (RPC)

1. `public.is_valid_wgs84_geometry(input_geometry geometry, allowed_types text[])` : Validates SRID 4326, non-empty status, geometric validity (`ST_IsValid`), allowed typemod, and WGS84 bounding limits.
2. `public.make_wgs84_bbox(min_lng, min_lat, max_lng, max_lat)` : Creates an `ST_MakeEnvelope` polygon with EPSG:4326 after validating coordinate ordering and bounding ranges.
3. `public.wgs84_distance_meters(origin, destination)` : Calculates geodetic surface distance in meters using PostGIS `geography` type casting.
4. `public.find_transport_nodes_near(origin geometry, radius_meters float8)` : Proximity search using `ST_DWithin(geometry::geography, ...)` ordered by distance.
5. `public.find_umkm_profiles_near(origin geometry, radius_meters float8)` : Proximity search for UMKM locations using `ST_DWithin` and geography GiST indexing.
6. `public.find_study_areas_within_bbox(min_lng, min_lat, max_lng, max_lat)` : Bounding box intersection using `&&` operator and `ST_Intersects` .
7. `public.find_transport_corridors_within_bbox(...)` : Bounding box query for transit lines.
8. `public.find_transport_nodes_within_bbox(...)` : Bounding box query for transit stations/nodes.
9. `public.find_umkm_profiles_within_bbox(...)` : Bounding box query for UMKM businesses.
10. `public.gettra_database_health()` : Lightweight RPC returning `'connected'` to verify database reachability.

NOTE

pgRouting Status: `NOT IMPLEMENTED / NOT VERIFIED` . pgRouting is not installed or enabled in the current database migrations. Shortest-path routing over pedestrian graphs remains a foundation contract pending graph data ingestion.

12. Data Model & Entities

GETRA core entity models are defined in `src/types/entity.ts` and `src/types/domain.ts` :

- **BaseEntity** : Base interface containing `id` , `created_at` , and `updated_at` .
 - **ProvenanceEntity** : Extends **BaseEntity** with `source_id` , `source_record_id` , `data_version` , `validation_status` , `retrieved_at` , `validated_at` , and `metadata` .
 - **StudyArea** : Represents polygon study zones.
 - **TransportCorridor** : Represents transit lines with `transport_mode` (`bus` , `rail` , `angkot` , `pedestrian` , etc.).
 - **TransportNode** : Represents transit stops with `node_type` (`station` , `stop` , `terminal`) and `transport_mode` .
 - **UkmProfile** : Represents local merchant points with `owner_id` , `business_name` , and `category` .
-

13. Repository Layer

The repository layer abstracts data access, utilizing the Supabase JavaScript Client with parameterization and strict schema mapping:

Repository Class	Location	Primary Responsibilities
<code>ProfileRepository</code>	<code>src/repositories/profile.repository.ts</code>	Profile retrieval and update via RLS
<code>StudyAreaRepository</code>	<code>src/repositories/study-area.repository.ts</code>	Study area querying and bounding box search
<code>TransportCorridorRepository</code>	<code>src/repositories/transport-corridor.repository.ts</code>	Corridor data access and bbox queries
<code>TransportNodeRepository</code>	<code>src/repositories/transport-node.repository.ts</code>	Node queries, proximity search, bbox search
<code>UmkmProfileRepository</code>	<code>src/repositories/umkm-profile.repository.ts</code>	Merchant profile management and spatial search
<code>SpatialSourceRepository</code>	<code>src/repositories/spatial-source.repository.ts</code>	Provenance source registry management
<code>ExternalRecordRepository</code>	<code>src/repositories/external-record.repository.ts</code>	Composition facade for external data persistence
<code>SupabaseHealthRepository</code>	<code>src/repositories/supabase-health.repository.ts</code>	Database connection probe (<code>getra_database_health</code>)
<code>SpatialRepository</code>	<code>src/modules/spatial/spatial.repository.ts</code>	Direct PostGIS RPC execution for distance/bbox/nearby

14. External Data Integration / MAPID Adapter

The external data integration module (`src/integrations/mapid/`) provides a clean pipeline for ingesting external GIS datasets without coupling GETRA core models to third-party proprietary schemas.

14.1 Status & Contract Verification

- **Integration Status:** FOUNDATION IMPLEMENTED – OFFICIAL CONTRACT REQUIRED
- **Real MAPID Connection:** NOT IMPLEMENTED (NO PRODUCTION INGESTION)
- **Design Pattern:** Inversion of Control with mockable HTTP transport, schema validation, and dry-run execution.

14.2 Adapter Pipeline Components

1. **MapidClient** (`src/integrations/mapid/mapid.client.ts`): HTTP client with timeout management (`AbortSignal.timeout`), retry logic, and credential masking.
 2. **MapidSchema** (`src/integrations/mapid/mapid.schema.ts`): Zod validation schemas for incoming external payload verification.
 3. **MapidNormalizer** (`src/integrations/mapid/mapid.normalizer.ts`): Transforms external attributes into standard WGS84 GeoJSON structures.
 4. **MapidMapper** (`src/integrations/mapid/mapid.mapper.ts`): Maps normalized data into repository-ready transport node entity inputs.
 5. **ExternalDataService** (`src/services/external-data/external-data.service.ts`): Orchestrates dry-run ingestion, duplicate detection, and provenance tracking.
-

15. Spatial Engine

The spatial engine (`src/modules/spatial/`) delivers core geospatial services backed by PostGIS:

- **DistanceService** : Computes accurate geodesic distance between two points in meters using PostGIS `ST_Distance(geometry::geography, destination::geography)` .
 - **ProximityService** : Performs bounded nearby searches (`ST_DWithin`) around a reference point with configurable radius (default max 50,000m).
 - **BBoxService** : Queries features intersecting a bounding box (`[west, south, east, north]`) with maximum span limits (default 10 degrees).
 - **WalkingTimeService** : Estimates pedestrian travel duration based on geodesic distance and configurable walking speed (default 1.4 m/s).
 - **ServiceAreaService** : Status: `FOUNDATION ONLY – PRODUCTION DATA REQUIRED` (Fails closed with `SPATIAL_NETWORK_NOT_READY`).
 - **RoutingService** / **UnavailableRoutingEngine** : Status: `NOT IMPLEMENTED` (Fails closed with `ROUTING_GRAPH_NOT_AVAILABLE`).
-

16. Docker Architecture

GETRA is containerized as a self-contained Next.js Standalone application.

16.1 Multi-Stage Dockerfile Lifecycle

- **Stage 1 (`deps`)**: Base `node:22-bookworm-slim`. Runs `npm ci` for deterministic dependencies.
- **Stage 2 (`builder`)**: Compiles application via `npm run build`, generating `.next/standalone` and `.next/static`.
- **Stage 3 (`runner`)**: Minimal production image. Copies standalone bundle, sets non-root user `node` (`USER node`), exposes port `3000`, and executes `node server.js`.

16.2 Container Healthcheck Specification

The container includes a native healthcheck probe:

BASH

```
HEALTHCHECK --interval=30s --timeout=5s --start-period=30s --retries=3 \
  CMD ["node", "-e", "fetch('http://127.0.0.1:3000/api/health',{signal:AbortSignal.timeout(4000)}).then(re
  sponse=>{if(!response.ok)process.exit(1)}).catch(()=>process.exit(1)}"]
```

17. Public API Security

GETRA implements comprehensive multi-layer security controls:

17.1 Security Controls Summary

- 1. CORS Allowlist:** Strict origin matching against `FRONTEND_ALLOWED_ORIGINS`. Wildcards (`*`), `null`, and malformed origins are rejected with `403 Forbidden`. Preflight `OPTIONS` handlers exist on every route.
 - 2. Rate Limiting:** In-memory fixed-window limiter per user/client IP:
 - **Auth Profile** (`auth:login`, `auth:register`): 5 requests / 60,000 ms.
 - **API Profile** (`/api/auth/me`, `/api/profile`): 60 requests / 60,000 ms.
 - **Mutation Profile** (`/api/auth/logout`, `PATCH /api/profile`): 20 requests / 60,000 ms.
 - **Spatial Profile** (`/api/spatial/*`): 30 requests / 60,000 ms.
 - 3. Bounded Request Body Reader:** Enforces strict byte caps on incoming JSON streams:
 - Global Default: 65,536 bytes (64 KB).
 - Auth Endpoints: 8,192 bytes (8 KB).
 - Profile & Spatial Endpoints: 4,096 bytes (4 KB).
 - 4. Security Headers:** Enforces `Content-Security-Policy: default-src 'none'`, `X-Frame-Options: DENY`, `X-Content-Type-Options: nosniff`, `Referrer-Policy: no-referrer`, `Permissions-Policy: camera=(), microphone=(), geolocation=()`, and production HSTS.
 - 5. No Service-Role Leakage:** Server routes strictly utilize publishable / user JWT client instances. No public API uses `SUPABASE_SERVICE_ROLE_KEY`.
-

18. API Catalog

The complete catalog of all 10 verified API endpoints available in GETRA:

#	Method	Endpoint Path	Classification	Auth Required	Rate Limit Profile	Purpose & Description
1	GET	/api/health	PUBLIC	None	None	Application readiness & Supabase connectivity probe
2	POST	/api/auth/login	PUBLIC	None	auth (5/min)	Authenticate user via email/password, return access token
3	POST	/api/auth/register	PUBLIC	None	auth (5/min)	Register new non-admin commuter, UMKM, or community user
4	POST	/api/auth/logout	AUTHENTICATED	Bearer JWT	mutation (20/min)	Acknowledge stateless client session termination
5	GET	/api/auth/me	AUTHENTICATED	Bearer JWT	api (60/min)	Get current authenticated user details and profile
6	GET	/api/profile	AUTHENTICATED	Bearer JWT	api (60/min)	Retrieve full user profile of the caller
7	PATCH	/api/profile	AUTHENTICATED	Bearer JWT	mutation (20/min)	Update user profile (<code>display_name</code> , <code>avatar_url</code>)
8	POST	/api/spatial/distance	AUTHENTICATED	Bearer JWT	spatial (30/min)	Compute geodesic distance in meters between two points
9	GET	/api/spatial/nearby	AUTHENTICATED	Bearer JWT	spatial (30/min)	Find nearby transport nodes or UMKM within radius
10	GET	/api/spatial/bbox	AUTHENTICATED	Bearer JWT	spatial (30/min)	Query spatial features within a bounding box

19. API Request & Response Examples

19.1 Health Check (`GET /api/health`)

Request:

```
HTTP

GET /api/health HTTP/1.1
Host: localhost:3000
x-request-id: 11111111-2222-3333-4444-555555555555
```

Success Response (HTTP 200 OK):

```
JSON

{
  "success": true,
  "data": {
    "service": "getra-api",
    "status": "ok",
    "database": "connected"
  },
  "request_id": "11111111-2222-3333-4444-555555555555"
}
```

Error Response — Database Down (HTTP 503 Service Unavailable):

```
JSON

{
  "success": false,
  "error": {
    "code": "DATABASE_UNAVAILABLE",
    "message": "Database connection failed",
    "retryable": true
  },
  "request_id": "11111111-2222-3333-4444-555555555555"
}
```

19.2 User Login (`POST /api/auth/login`)

Request:

```
HTTP

POST /api/auth/login HTTP/1.1
Host: localhost:3000
Content-Type: application/json

{
  "email": "commuter.test@example.com",
  "password": "ExamplePassword123!"
}
```

Success Response (HTTP 200 OK):

JSON

```
{
  "success": true,
  "data": {
    "user": {
      "id": "e8d0e512-1823-455b-8025-a1312384a601",
      "email": "commuter.new@example.com"
    },
    "profile": {
      "display_name": "Budi Commuter",
      "role": "COMMUTER"
    }
  },
  "request_id": "4d983e20-7ad1-49b8-bc88-34827fb70588"
}
```

Error Response — Admin Role Attempt (HTTP 403 Forbidden):

JSON

```
{
  "success": false,
  "error": {
    "code": "FORBIDDEN",
    "message": "Role ADMIN tidak dapat dibuat melalui public registration.",
    "retryable": false
  },
  "request_id": "4d983e20-7ad1-49b8-bc88-34827fb70588"
}
```

19.4 Spatial Distance Calculation (`POST /api/spatial/distance`)

Request:

HTTP

```
POST /api/spatial/distance HTTP/1.1
Host: localhost:3000
Authorization: Bearer <VALID_JWT_TOKEN>
Content-Type: application/json
```

```
{
  "origin": {
    "longitude": 107.60981,
    "latitude": -6.914744
  },
  "destination": {
    "longitude": 107.61861,
    "latitude": -6.90389
  }
}
```

Success Response (HTTP 200 OK):

JSON

```
{
  "success": true,
  "data": {
    "distance_meters": 1542.85,
    "origin": {
      "longitude": 107.60981,
      "latitude": -6.914744
    },
    "destination": {
      "longitude": 107.61861,
      "latitude": -6.90389
    },
    "analysis_method": "postgis_geography_distance",
    "source": "postgis_st_distance",
    "srid": 4326
  },
  "request_id": "5c983e20-7ad1-49b8-bc88-34827fb70599"
}
```

19.5 Spatial Nearby Query (GET /api/spatial/nearby)

Request:

HTTP

```
GET /api/spatial/nearby?lat=-6.914744&lng=107.60981&radius=1000&type=transport_node&limit=10 HTTP/1.1
Host: localhost:3000
Authorization: Bearer <VALID_JWT_TOKEN>
```

Success Response (HTTP 200 OK):

JSON

```
{
  "success": true,
  "data": {
    "origin": {
      "longitude": 107.60981,
      "latitude": -6.914744
    },
    "radius_meters": 1000,
    "type": "transport_node",
    "features": [
      {
        "id": "10000000-0000-0000-0000-000000000004",
        "entity_type": "transport_node",
        "label": "Halte Alun-Alun Bandung",
        "geometry": {
          "type": "Point",
          "coordinates": [107.60985, -6.9148]
        },
        "provenance": {
          "source_id": "10000000-0000-0000-0000-000000000001",
          "source_name": "TEST MANUAL SOURCE",
          "source_type": "manual"
        }
      }
    ],
    "returned_count": 1,
    "analysis_method": "postgis_st_dwithin",
    "source": "postgis_rpc",
    "srid": 4326
  },
  "request_id": "6d983e20-7ad1-49b8-bc88-34827fb70500"
}
```

19.6 Spatial Bounding Box Query (`GET /api/spatial/bbox`)

Request:

HTTP

```
GET /api/spatial/bbox?west=107.60&south=-6.95&east=107.65&north=-6.90&type=umkm_profile&limit=20 HTTP/1.1
Host: localhost:3000
Authorization: Bearer <VALID_JWT_TOKEN>
```

Success Response (HTTP 200 OK):

JSON

```
{
  "success": true,
  "data": {
    "bbox": {
      "west": 107.6,
      "south": -6.95,
      "east": 107.65,
      "north": -6.9
    },
    "type": "umkm_profile",
    "features": [
      {
        "id": "10000000-0000-0000-0000-000000000005",
        "entity_type": "umkm_profile",
        "label": "Warung Kopi Braga",
        "geometry": {
          "type": "Point",
          "coordinates": [107.609, -6.917]
        },
        "provenance": {
          "source_id": "10000000-0000-0000-0000-000000000001",
          "source_name": "TEST MANUAL SOURCE",
          "source_type": "manual"
        }
      }
    ],
    "returned_count": 1,
    "analysis_method": "postgis_bbox_intersection",
    "source": "postgis_rpc",
    "srid": 4326
  },
  "request_id": "7d983e20-7ad1-49b8-bc88-34827fb70511"
}
```

20. Authentication Examples Per Role

Registration requests are customized per role. All roles use the identical `POST /api/auth/login` endpoint for authentication, with subsequent permissions governed by the user's assigned role in `public.profiles`.

20.1 Commuter Registration

JSON

```
{
  "email": "commuter.user@example.com",
  "password": "ExamplePassword123!",
  "display_name": "Ahmad Commuter",
  "role": "COMMUTER"
}
```

20.2 UMKM Merchant Registration

JSON

```
{
  "email": "umkm.merchant@example.com",
  "password": "ExamplePassword123!",
  "display_name": "Kedai Kopi Transit",
  "role": "UMKM"
}
```

20.3 Community Representative Registration

JSON

```
{
  "email": "community.rep@example.com",
  "password": "ExamplePassword123!",
  "display_name": "Komunitas Pejalan Kaki",
  "role": "COMMUNITY"
}
```

21. Supabase Migration Runbook

The project uses Supabase CLI for database schema versioning.

Project Reference ID: `sesakxnjaphrxqx11qjm`

21.1 Standard Migration Workflow

MERMAID

```
graph TD
    A[Schema Change Required] --> B[Create New Migration File: supabase migration new name]
    B --> C[Write SQL DDL / Constraints / RLS / Indexes]
    C --> D[Preview Changes: supabase db push --dry-run]
    D --> E{Dry-Run OK?}
    E -- No --> C
    E -- Yes --> F[Apply to Remote: supabase db push]
    F --> G[Verify Status: supabase migration list]
```

21.2 CLI Terminal Commands Guide

Step 1: Verify CLI Version & Login

BASH

```
supabase --version
supabase login
```

Step 2: Link Workspace to Remote Project

BASH

```
supabase link --project-ref sesakxnjaphrxqx11qjm
```

Step 3: Check Current Migration Status

BASH

```
supabase migration list
```

Step 4: Create a New Migration File

BASH

```
supabase migration new add_feature_table_name
```

Step 5: Dry-Run Preview Before Applying

BASH

```
supabase db push --dry-run
```

Step 6: Push Migration to Remote Database

BASH

```
supabase db push
```

Step 7: Verify Applied Migrations

BASH

```
supabase migration list
```

CAUTION

Destructive Operations Policy: Never execute `supabase db reset --linked` on remote or production environments. Destructive commands (`DROP TABLE` , `DROP COLUMN`) require manual architectural review and forward-migration scripts.

22. Database Seeding Guide

22.1 Seeding Structure

The file `supabase/seed.sql` contains **strictly synthetic development fixtures** prefixed with `TEST ...`.

- Contains mocked users (`admin.mock@getra.local` , `umkm.mock@getra.local` , `commuter.mock@getra.local`).
 - Contains synthetic bounding boxes, test lines, and mock coordinates.
 - **Strict Rule:** `seed.sql` is intended exclusively for local Docker/Supabase testing (`supabase db reset`). It must **never** be applied to production.
-

23. Database Types Generation

When linked to the Supabase remote project, TypeScript database definitions can be generated to keep frontend and backend types synchronized:

```
BASH
```

```
# Generate types directly from remote schema  
npx supabase gen types typescript --project-ref sesakxnjaphrxqxllqjm > src/types/database.types.ts
```

24. Development Mode Running Guide

For local feature development, interactive debugging, and hot-reloading:

BASH

```
# 1. Install dependencies
npm install

# 2. Run TypeScript type check
npm run typecheck

# 3. Run ESLint
npm run lint

# 4. Start Next.js development server (default port 3000)
npm run dev
```

NOTE

`npm run dev` is strictly for local interactive development and debugging. It must **not** be used as a 24/7 background production server.

25. Docker Container Running Guide

25.1 Build Container Image

BASH

```
# Using npm script
npm run docker:build

# Or directly using Docker CLI
docker build -t getra-backend:latest .
```

25.2 Start Background Container (Detached Mode)

BASH

```
# Start container in detached mode (-d)
npm run docker:start

# Equivalent Compose command
docker compose --env-file .env.local up -d
```

25.3 Check Container Status

BASH

```
npm run docker:status

# Equivalent Compose command
docker compose --env-file .env.local ps
```

25.4 View Real-time Container Logs

BASH

```
# Follow live logs
npm run docker:logs

# Or using Compose directly
docker compose --env-file .env.local logs -f --tail 100
```

25.5 Restart Container

BASH

```
npm run docker:restart
```

25.6 Stop Container

BASH

```
npm run docker:stop
```

26. 24/7 Production Operations Guide

For continuous 24/7 production operation, GETRA backend must run inside Docker with automatic restart policies or under PM2 process supervision.

26.1 24/7 Infrastructure Architecture



26.2 Automatic Crash Restart Behavior

- Under Docker Compose (`restart: unless-stopped`), if the Node.js application experiences an unhandled exception or process termination, the Docker daemon automatically instantiates a new container instance.
- Under PM2 (`max_restarts: 10` , `autorestart: true`), crashed processes are automatically rebooted within `2000ms` .

27. Backend Logs & Monitoring

27.1 Structured Logging Format

GETRA implements sanitized JSON-formatted console logging via `withApiLogger`:

JSON

```
{
  "timestamp": "2026-08-17T15:50:00.000Z",
  "level": "info",
  "service": "getra-backend",
  "request_id": "4d983e20-7ad1-49b8-bc88-34827fb70588",
  "method": "POST",
  "path": "/api/spatial/distance",
  "status": 200,
  "duration_ms": 14
}
```

27.2 Error Sanitization

All database errors, Supabase exceptions, and internal failures are intercepted. Stack traces, database hostnames, usernames, and raw SQL queries are stripped before response transmission.

28. Frontend Integration Guide

Frontends (Next.js web apps, mobile apps) interact with GETRA via standard JSON HTTP requests.

28.1 Frontend Client Request Example (TypeScript)

TYPESCRIPT

```
interface ApiResponse<T> {
  success: boolean;
  data?: T;
  error?: {
    code: string;
    message: string;
    retryable: boolean;
  };
  request_id: string;
}

export async function fetchSpatialNearby(
  token: string,
  lat: number,
  lng: number,
  radius: number = 1000
): Promise<ApiResponse<any>> {
  const baseUrl = process.env.NEXT_PUBLIC_GETRA_API_URL || "http://localhost:3002";

  const response = await fetch(
    `${baseUrl}/api/spatial/nearby?lat=${lat}&lng=${lng}&radius=${radius}&type=transport_node`,
    {
      method: "GET",
      headers: {
        "Authorization": `Bearer ${token}`,
        "Accept": "application/json",
      },
    }
  );

  return response.json();
}
```

29. Environment Variables Reference

Variable Name	Required	Target Scope	Purpose / Default Value	Secret Level
NEXT_PUBLIC_SUPABASE_URL	YES	Client & Server	Supabase project API URL (<a href="https://<ref>.supabase.co">https://<ref>.supabase.co)	Public
NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY	YES	Client & Server	Supabase anonymous public API key	Public
APP_ENV	Optional	Server-only	Application environment (<code>development</code> , <code>staging</code> , <code>production</code>)	Safe Config
APP_BASE_URL	Optional	Server-only	Public HTTP/HTTPS origin (e.g. <code>http://localhost:3002</code>)	Safe Config
FRONTEND_ALLOWED_ORIGINS	Optional	Server-only	Comma-separated CORS allowlist (e.g. <code>http://localhost:3000</code>)	Safe Config
TRUST_PROXY	Optional	Server-only	Trust reverse proxy headers (<code>true</code> or <code>false</code>)	Safe Config
API_MAX_JSON_BODY_BYTES	Optional	Server-only	Global JSON body stream cap (default: <code>65536</code> bytes)	Safe Config
SUPABASE_REQUEST_TIMEOUT_MS	Optional	Server-only	Supabase fetch abort timeout (default: <code>10000</code> ms)	Safe Config
RATE_LIMIT_WINDOW_MS	Optional	Server-only	Fixed window duration (default: <code>60000</code> ms)	Safe Config
RATE_LIMIT_AUTH_MAX_REQUESTS	Optional	Server-only	Auth endpoint limit (default: <code>5</code> req/window)	Safe Config
RATE_LIMIT_API_MAX_REQUESTS	Optional	Server-only	API endpoint limit (default: <code>60</code> req/window)	Safe Config
RATE_LIMIT_MUTATION_MAX_REQUESTS	Optional	Server-only	Mutation endpoint limit (default: <code>20</code> req/window)	Safe Config
RATE_LIMIT_SPATIAL_MAX_REQUESTS	Optional	Server-only	Spatial endpoint limit (default: <code>30</code> req/window)	Safe Config
SPATIAL_MAX_RADIUS_METERS	Optional	Server-only	Maximum spatial search radius (default: <code>50000</code> m)	Safe Config
SPATIAL_MAX_BBOX_LATITUDE_DEGREES	Optional	Server-only	Max bbox latitude span (default: <code>10.0</code> deg)	Safe Config
SPATIAL_MAX_BBOX_LONGITUDE_DEGREES	Optional	Server-only	Max bbox longitude span (default: <code>10.0</code> deg)	Safe Config
DEFAULT_WALKING_SPEED_MPS	Optional	Server-only	Pedestrian speed estimator (default: <code>1.4</code> m/s)	Safe Config
SUPABASE_SERVICE_ROLE_KEY	Optional	Server-only	Administrative Supabase key (Reserved / Backend-only)	CRITICAL SECRET
MAPID_API_KEY	Optional	Server-only	External provider API key (Reserved / Backend-only)	CRITICAL SECRET

30. Security Checklist & Compliance

- [x] **Row Level Security:** Enabled and verified on all 6 database tables.
 - [x] **Column-Level Permissions:** Role modification restricted from user profiles; provenance tampering blocked on UMKM tables.
 - [x] **Admin Role Protection:** Public registration strictly rejects `ADMIN` creation at both database trigger and application route layers.
 - [x] **CORS Enforcement:** Exact origin matching; wildcard `*` and null origins rejected with HTTP 403.
 - [x] **Rate Limiting:** Active in-memory fixed-window limiter enforcing 5–60 req/min depending on route criticality.
 - [x] **Payload Bounds:** Strict byte limits on all incoming request JSON streams.
 - [x] **Credential Isolation:** `.env*` excluded from Docker build context; zero hardcoded secrets in repository.
 - [x] **Container Non-Root User:** Dockerfile explicitly specifies `USER node`.
 - [x] **Error Sanitization:** Zero database passwords, connection strings, or stack traces leaked in error responses.
-

31. Testing & Quality Assurance Guide

The repository includes a comprehensive 44-file, 353-test automated test suite:

BASH

```
# 1. Run all unit and integration tests
npm test

# 2. Run TypeScript type checking
npm run typecheck

# 3. Run ESLint code quality verification
npm run lint

# 4. Compile production build
npm run build
```

32. Troubleshooting & Diagnostics Guide

Problem / Error	Root Cause	Solution
<code>npm run build</code> fails	Missing environment variables or TypeScript error	Run <code>npm run typecheck</code> to locate type mismatch. Ensure <code>.env.local</code> contains <code>NEXT_PUBLIC_SUPABASE_URL</code> .
<code>LegacyProjectNotLinkedError</code>	Supabase CLI has not linked local workspace	Run <code>supabase link --project-ref sesakxnjaphrxqxllqjm</code> and authenticate.
HTTP <code>401 Unauthorized</code>	Missing or invalid Supabase Bearer token	Verify client transmits <code>Authorization: Bearer <valid_jwt></code> header.
HTTP <code>403 Forbidden</code> (<code>CORS_ORIGIN_DENIED</code>)	Origin not in allowlist	Add frontend origin to <code>FRONTEND_ALLOWED_ORIGINS</code> in <code>.env.local</code> .
HTTP <code>429 Too Many Requests</code>	Rate limit threshold exceeded	Client must wait and respect <code>Retry-After</code> response header.
HTTP <code>503 Service Unavailable</code>	Supabase unreachable or health check failed	Check outbound internet connection, Supabase service status, and URL credentials.
Docker Daemon Error (HTTP 500)	Docker Desktop service not running on host	Start Docker Desktop application or start the Windows Docker service.

33. Known Limitations

1. **In-Memory Rate Limiter:** Current rate limiting is process-local. Distributed deployment across multiple instances requires migrating to a shared Redis store.
 2. **Pedestrian Routing Graph:** No production pedestrian graph is active; routing service responds with `ROUTING_GRAPH_NOT_AVAILABLE`.
 3. **Official MAPID Contract:** Live MAPID integration is blocked pending verified production contracts and API credentials.
 4. **Token Revocation:** Logout is client-side stateless token discarding; global token blacklisting is not implemented.
-

34. Features Waiting for Production Data

Feature Name	Current Implementation Status	Required Production Data Dependency
Production Pedestrian Routing	FOUNDATION ONLY	Complete, routable topological pedestrian road graph
Isochrone Service Area Analysis	FOUNDATION ONLY	Multimodal transit timetable and network graph data
MAPID Live Sync & Ingestion	FOUNDATION ONLY	Verified MAPID API contracts, endpoints, and credentials
Demand Intelligence Platform	NOT IMPLEMENTED	Mobile commuter movement telemetry & survey data
Fair Discovery Transit Matching	NOT IMPLEMENTED	Real-time transit arrival feeds (GTFS-RT / GPS)

35. Phase Completion Summary (Phases 0–7 + Extra)

Phase	Title	Implementation Status	Main Deliverables	Database Changes
Phase 0	Development Foundation	COMPLETED	Toolchain, Next.js 16 setup, linting, Vitest	None
Phase 1	Authentication	COMPLETED	Supabase Auth login, register, me, logout routes	None
Phase 2	Profile, Role & RLS	COMPLETED	Profile schema, role guards (COMMUTER , UMKM , etc.)	profiles table
Phase 3	API Foundation	COMPLETED	Standard envelope, error hierarchy, logging, validation	None
Phase 4	Database & PostGIS Foundation	COMPLETED	5 core spatial tables, GiST indexes, WGS84 functions	5 spatial tables
Phase 5	Data Model & Repository Layer	COMPLETED	Provenance tracking, repository contracts, immutable triggers	Migration update
Phase 6	MAPID / External Data Adapter	COMPLETED	Adapter, client, normalizer, fixture test flow	None
Phase 7	Spatial Engine Foundation	COMPLETED	Distance, Nearby, BBox APIs, walking time estimator	PostGIS RPCs
Phase Extra	Docker & Public API Security	COMPLETED	Standalone Dockerfile, CORS, rate limiting, request caps	None
Final	Master Documentation & Handover	COMPLETED	Master PDF manual, complete API catalog, audit	None

36. Operational Handover Checklist

- [x] Full repository audit completed.
 - [x] 10 verified API endpoints cataloged with exact JSON payloads.
 - [x] 6 database tables, constraints, and GiST indexes documented.
 - [x] RLS policies and role authorization matrix verified.
 - [x] PostGIS foundation and RPC functions documented.
 - [x] Supabase migration and seeding runbooks created.
 - [x] Development and Docker 24/7 deployment procedures verified.
 - [x] Frontend integration code samples provided.
 - [x] Security checklist and troubleshooting matrix completed.
 - [x] Secret audit passed with zero credential leakage.
 - [x] Final PDF and Markdown documentation generated in `docs/final/`.
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GETRA Platform Technical Documentation — Extra Phase Final Handover Complete.